GGG-S-116b
APRIL 5, 1961
SUPERSEDING
Fed. Spec. GGG-S-116a
February 15, 1952

FEDERAL SPECIFICATION

# SCRAPERS, SHIP, (PAINT AND SCALE REMOVING)

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

1.1 Scope.—Scrapers covered by this specification are hand tools for removing paint and scale from finished surfaces.

1.1.1 *Federal Specification coverage.* — Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are ordinarily used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* — Ship scrapers shall be of the following types and classes, as specified (see 6.1):

Type I.—All-metal:
Class A.—Angle-bent end.
Class B.—Curved-bent end.
Type II.—Deck, wood-handle.
Type III.—Hull, wood-handle.
Type IV.—Ship-bottom, wood-handle.
Type V.—Ships, triangular, wood-handle.

## 2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

*Federal Standards:*

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

*Military Specifications:*

MIL-P-15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

*Military Standards:*

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.



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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 The illustrations herein are descriptive and not restrictive, except for the dimensions, and are not intended to preclude the purchase of ship scrapers which are otherwise in accordance with the requirements of this specification.

**3.2 Material.**

3.2.1 *Steel.* — Steel shall be as specified hereinafter for each type.

3.2.2 *Wood handles.*—Wood handles shall be of good-quality straight-grained hickory, hard maple, or tough white ash.

**3.3 Hardness.**—At a midpoint within ½ inch back from scraping edges, the blade shall show a hardness value of not less than 57 nor more than 63 on the Rockwell C scale, when tested as described in 4.5.2.

**3.4 Identification marking.** — Each ship scraper shall be stamped or otherwise permanently marked in a plain and legible manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

**3.5 Finish.**

3.5.1 *Steel.* — Steel shall have either a painted, lacquered, or natural finish, at the option of the contractor, except for the scraping edges which shall be ground. Unless otherwise specified, all surfaces having a natural or ground finish shall be completely coated on delivery with an oil or grease compound suitable for the prevention of rust or corrosion in storage without adversely affecting the material coated.

3.5.2 *Wood handles.* — Unless otherwise specified in the invitation for bids, wood handles shall be either with or without a protective coating of varnish, lacquer, shellac, or wax, at the option of the contractor.

**3.6 Type I, class A, all-metal scrapers, angle-bent end.**—Type I, class A scrapers shall be generally in accordance with figure 1.

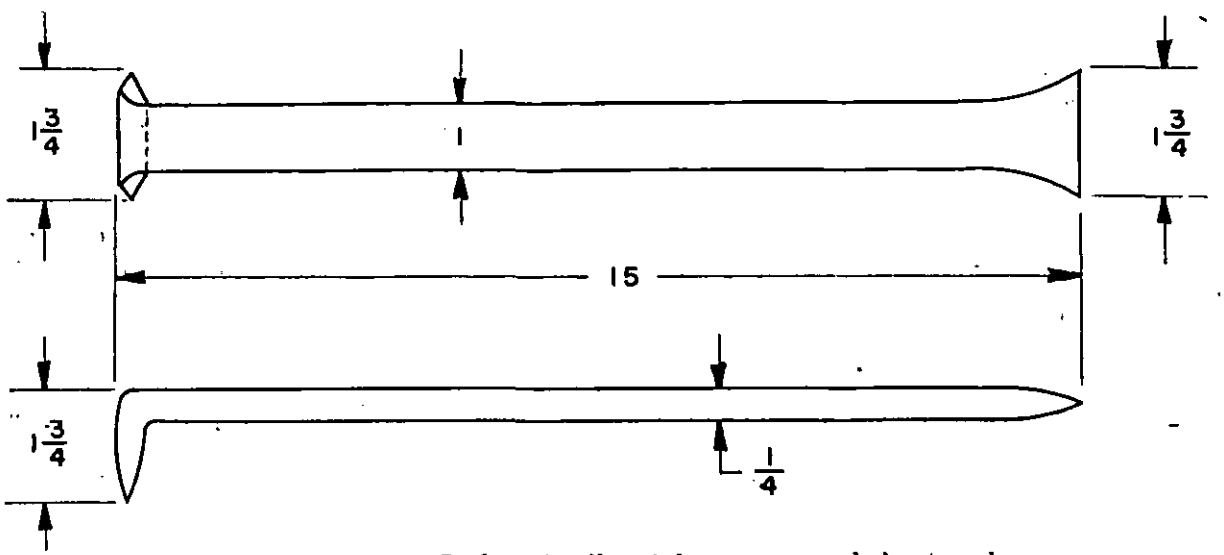



FIGURE 1.—Type I, class A, all-metal scraper, angle-bent end.



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3.6.1 *Manufacture and design.* — The scrapers shall be made from one piece of steel having a carbon content of 0.90 to 1.05 percent. Both ends of the tool shall have scraping edges, one of which shall be in alinement with, and the other at approximately right angles to, the longitudinal axis of the scraper body. Both ends shall be so forged, hardened, tempered, and ground to a double-beveled scraping edge, that the product shall be of fine silky grain structure and shall be satisfactory for the purpose intended.

3.6.2 *Dimensions.*—The scrapers shall comply with the dimensions given on figure 1 except that the dimensions may vary within a tolerance of plus or minus 5 percent.

3.7 Type I, class B, all-metal scrapers, curved-bent end.—Type I, class B scrapers shall be generally in accordance with figure 2.

3.7.1 *Manufacture and design.* — The scrapers shall comply with the requirements of 3.6.1 except that the two scraping edges shall be ground to a single-beveled edge.

3.7.2 *Dimensions.*—The scrapers shall comply with the dimensions given on figure 2 except that the dimensions may vary within a tolerance of plus or minus 5 percent.

3.8 Type II, deck scrapers, wood-handle.—Type II scrapers shall be generally in accordance with figure 3.

3.8.1 *Manufacture and design.* — The scrapers shall have a blade made from steel having a carbon content of 0.80 to 0.90 per-

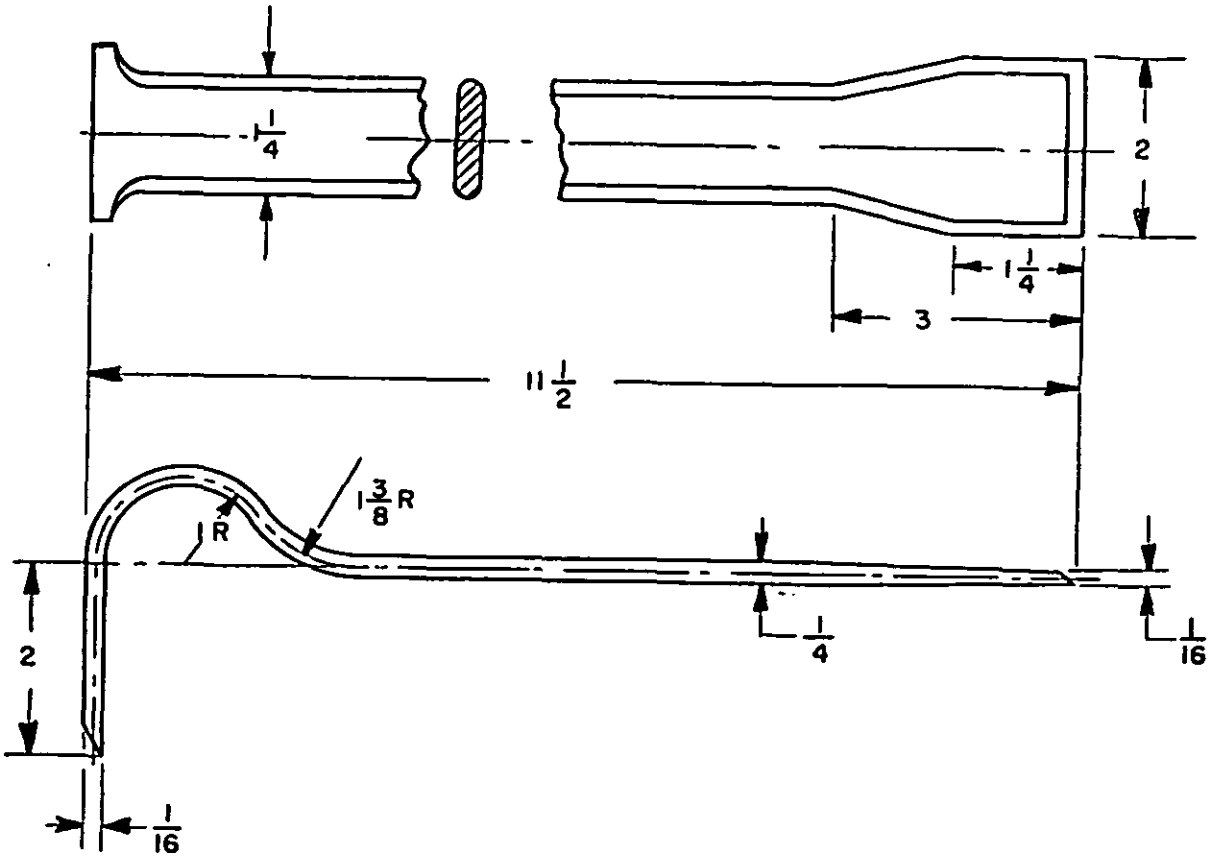



FIGURE 2.—Type I, class B, all-metal scraper, curved-bent end.



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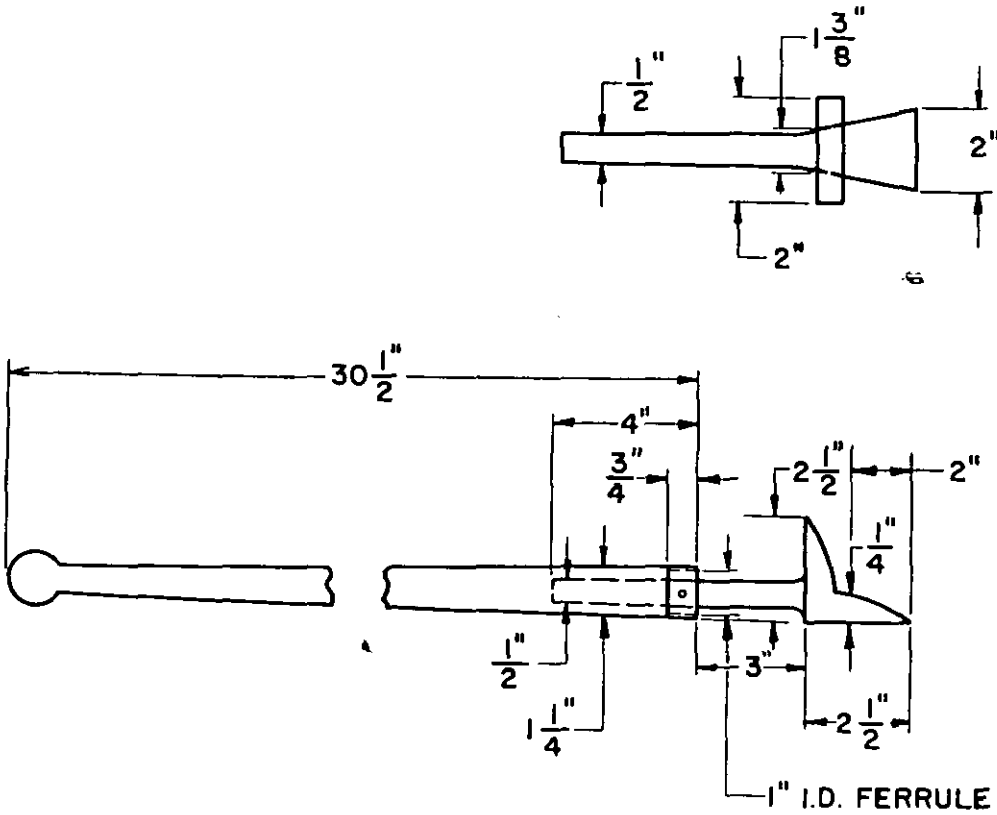



FIGURE 3.—Type II, deck scraper, wood handle.

cent. The blade shall have two scraping edges located 90 degrees apart and properly hardened, tempered, and ground; one of the edges shall be in longitudinal alignment with the handle and the other edge at approximately right angles to the handle. The blade shall be either forged integrally with, or securely fastened to, a steel shank which shall fit securely into a wood handle. A substantial metal ferrule and a through rivet shall be provided to hold the shank securely in the handle under service conditions.

3.8.2 *Handle.*—The handle shall be well proportioned and of suitable shape and size to provide a comfortable handgrip.

3.8.3 *Dimensions.*—The scrapers shall comply with the dimensions given in figure 3 except that the dimensions may vary within a tolerance of plus or minus 5 percent.

**3.9 Type III, hull scrapers, wood-handle.**—(See figure 4.)

3.9.1 *Manufacture and design.*—Type III scrapers shall be as specified for type II deck scrapers except that the handle length shall be 18 inches, subject to a tolerance of plus or minus 5 percent.

**3.10 Type IV, ship-bottom scrapers, wood handle.**—(See figure 5.)

3.10.1 *Manufacture and design.* — The scrapers shall have a blade made of steel having a carbon content of 0.80 to 0.90 percent. The blade shall be properly hardened and tempered, and the scraping edge shall be ground to a double-beveled chisel edge. The blade shall rest in a slot and shall be secured by at least three rivets to a steel or malleable-iron socket. The socket shall be provided with



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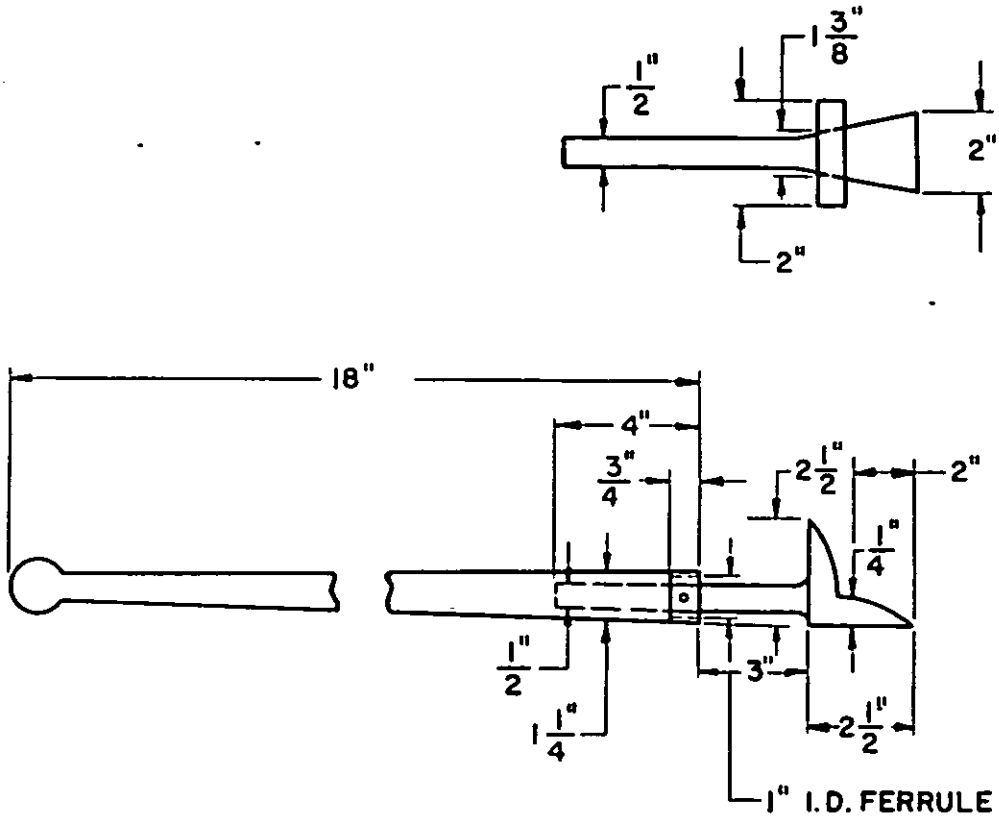



FIGURE 4.—Type III, hull scraper, wood handle.

a taper bore into which the turned taper end of the handle shall be neatly fitted and secured by means of a number 10, 7/8-inch-long, steel, wood screw.

3.10.2 *Handle.*—The handle shall be well proportioned and of a suitable size and shape to provide a comfortable handgrip.

3.10.3 *Dimensions.* — The scrapers shall comply with the dimensions given on figure 4 except that the dimensions may vary within a tolerance of plus or minus 5 percent.

**3.11 Type V, ships triangular scrapers, wood-handle.**—(See figure 6.)

3.11.1 *Manufacture and design.* — Type V scrapers shall consist of a blade, a socket, and a wood handle.

3.11.1.1 *Blade.*—The blade shall be made of steel having a carbon content of 0.80 to 0.90 percent. It shall be trapezium-shaped, having four scraping edges, each ground at an angle of approximately 60 degrees, and shall be provided with a rectangular hole approximately 3/8 by 7/8 inch in the center. The thickness of the blade shall be about 1/4 inch, the lengths of the sides shall be approximately 4 3/4, 4 3/4, 4 inches, and 3/4 inch, respectively. The blade shall be secured to the socket by means of a bolt and washer. The bolt shall be 3/4 inch in length, threaded 3/8-inch American National form, and the head shall be either square or hexagonal.

3.11.1.2 *Socket.*—The socket shall be of malleable iron, galvanized, and furnished with a straight-grained, well-seasoned hickory handle, which shall be inserted in the socket and secured to it by means of three or more round-head brass screws. The socket shall conform to the following approximate dimensions: Length overall, 5 1/2 inches; out-



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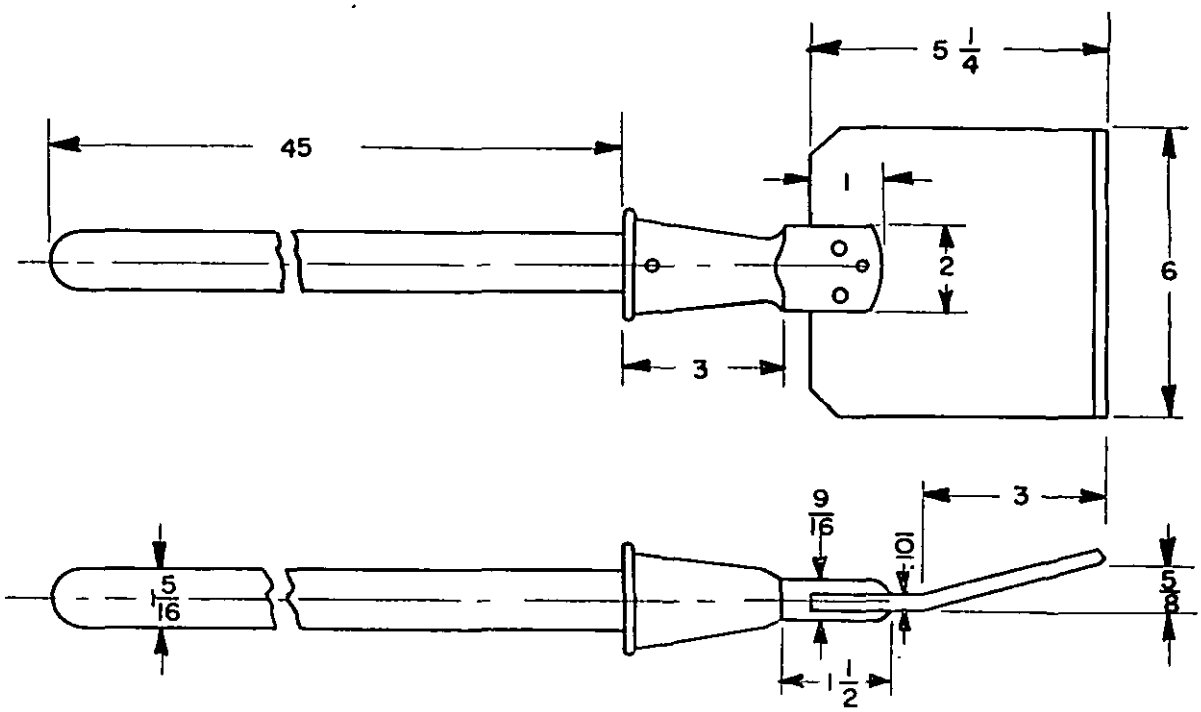



FIGURE 5.—Type IV, ship-bottom scraper, wood handle.

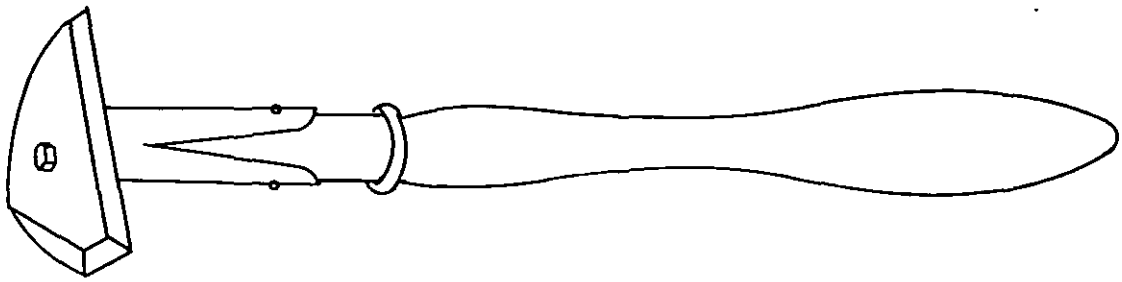


FIGURE 6.—Type V, ships, triangular scraper, wood handle.

side diameter large end, 1¼ inches; small end, ¾ inch. The small end of the socket shall have a rectangular shoulder about 3/16 inch in height which shall fit the hole in the blade. The shoulder of the socket shall be tapped ⅜ American National tread to suit a ⅜ by ¾-inch bolt.

3.11.1.3 *Handle.*—The wood handle shall be tapered to fit the socket and shall be inserted for a distance of about 4 inches. The large diameter of the handle shall be approximately 1¼ inches; the overall length of the complete scraper shall be about 18 inches.

**3.12 Workmanship.**

3.12.1 *Steel.*—The steel shall be free from burrs, roughness, or excessive oxide scale.

3.12.2 *Wood handles.*—The wood handles shall be smoothly finished.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise speci-



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fied, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Lot.**—All scrapers of the same type, class, and size, presented at one time, shall be considered a lot for purposes of acceptance inspection.

4.3 Sampling for lot acceptance.

4.3.1 *Sampling for examination.*

4.3.1.1 *Sampling for intermittent procurement in isolated lots.*—A random sample of scrapers shall be taken from each inspection lot of material offered for examination with lot acceptance based on the single sampling-inspection requirements shown in table I, in accordance with Military Standard MIL–STD–105.

TABLE I.—*Sampling for examination AQL (approx.) = 2.5 percent defective*

| Number of scrapers in lot | Number of scrapers in sample | Acceptance number (defectives) | Rejection number (defectives) |
|---|---|---|---|
| 15 and under | 7 | 0 | 1 |
| 16 to 40 | 10 | 0 | 1 |
| 41 to 110 | 15 | 1 | 2 |
| 111 to 300 | 25 | 1 | 2 |
| 301 to 500 | 35 | 2 | 3 |
| 501 to 800 | 50 | 3 | 4 |
| 801 to 1,300 | 75 | 4 | 5 |
| 1,301 and over | 110 | 6 | 7 |

4.3.1.2 *Sampling for continuous production.*—Where there is continuous production for the Government and lots include at least 200 items, sampling for inspection shall be conducted in accordance with Military Standard MIL–STD–105 at inspection level I and AQL = 2.5 percent defective; the type of sampling to be determined by the inspector.

4.3.2 *Sampling for tests.*—A random sample of at least 4 scrapers shall be taken from each lot of 1,000 or fraction thereof and shall be subjected to the tests specified in 4.5.

4.4 **Examination.** — Each of the sample scrapers taken in accordance with 4.3.1 shall be examined to verify compliance with the requirements of this specification. Examination shall be conducted as specified in table II. Any scraper in the sample containing one or more defects shall be rejected, and if the number of defective scrapers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Tests.

4.5.1 *Carbon content of steel.*—At least 2 ounces of drillings shall be obtained from at least two of the four samples taken for each type and class for this test. Analysis shall be made of the drillings to determine compliance with the specified requirements for carbon content. All sample drillings shall be clean, fine, dry, and free from foreign substances. The drillings shall be taken at slow speed and shall not be oxidized. If surfaces are contaminated by foreign matter or are otherwise not representative of the quality of the steel tested, surface chips shall be discarded. Where required to facilitate drilling, the temper may be drawn. When satisfactory to the inspector, a certificate of analysis may be accepted in lieu of making the analysis of the drillings.

4.5.2 *Hardness test.*—At least two of the samples taken for each type and class shall be tested on a Rockwell hardness testing machine, employing a 150-kilogram major load and using a standard 120-degree diamond



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TABLE II.—*Classification of defects in accordance with Military Standard MIL–STD–105*

| Categories | Defects |
|---|---|
| *Critical:* | |
| 1 | None defined. |
| *Major:* | |
| 101 | Scrapers not type, class specified. |
| 102 | Evidence of use of unauthorized materials. |
| 103 | Wood handles not straight grain hickory, maple or ash as specified; not well proportioned; not a comfortable grip; not suitable shape. |
| 104 | Wood handles not ferruled; ends not tapered; not diameter or length specified; not inserted in socket specified distance. |
| 105 | Configuration and dimensions not in conformance with approved figures. |
| 106 | Blades not shape, size specified; not secured to socket; edges not ground; securing bolt threads nonconforming. |
| *Minor:* | |
| 201 | Scrapers not marked as specified; marking not permanent; not legible; data incomplete. |
| 202 | Steel not painted, finished, or preserved as required. |
| 203 | Wood handles not covered with protective coating where specified. |
| 204 | Steel not free of burrs, rough spots, oxide scale; handles not smooth finish. |

penetrator. Scrapers shall be prepared for hardness tests by grinding the two opposite sides flat and parallel, particularly to remove the effect of any casehardening. Rockwell hardness impressions shall be taken on these ground surfaces at a midpoint of the blades within ½ inch of the scraping edges, and the average result of three impressions shall comply with the requirements of 3.3.

4.5.3 *Grain structure.*—Both ends of type I ship scraper samples shall be nicked and broken approximately parallel to and at a distance of ¼ to ⅜ inch back from the scraping edges and shall be examined for conformance with the requirements of 3.6.1.

4.5.4 *Possible test failures.*—Possible test failures are defined as follows:

(*a*) Metallurgical examination:
Carbon content of steel not in accordance with specification.
Grain structure nonconforming (type I, class A).

(*b*) Hardness tests:
Test samples not within limiting Rockwell "C" range.

## 5. PREPARATION FOR DELIVERY

(Civil agencies.—Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

**5.1 Cleaning, preservation, and packaging.**

5.1.1 *Level A.*—Scrapers shall be preserved and packaged in accordance with level A of Military Specification MIL–P–15424.

5.1.2 *Level C.*—Scrapers shall be preserved and packaged in accordance with manufacturer's commercial practice.

**5.2 Packing.**

5.2.1 *Level A.* — Scrapers, packaged as specified shall be packed in accordance with level A requirements of Military Specification MIL–P–15424.

5.2.2 *Level B.* — Scrapers, packaged as specified shall be packed in accordance with level B requirements of Military Specification MIL–P–15424.

5.2.3 *Level C.* — Scrapers, packaged as specified shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or other carrier regulations applicable to the mode of transportation.

**5.3 Marking.**

5.3.1 *Civil agencies.*—In addition to any special marking specified in the contract or



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order, each unit and intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military.*—In addition to any special marking required by the contract or order, or herein, interior and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

## 6. NOTES

6.1 Ordering data. — Procurement documents should specify the following and purchasers should exercise any desired options offered herein:

(*a*) Title, number, and date of this specification.
(*b*) Type and class as specified (see 1.2).
(*c*) Level of packaging and packing required (see 5.1 and 5.2).
(*d*) Marking, if different (see 5.3).

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with the requirements of this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically called for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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